AI-Driven Reversal of the C4A Transcriptomic Axis Reveals Novel Therapeutic Drug Candidates for Schizophrenia

Abstract

Schizophrenia's genetic risk signal C4A drives pathological synaptic pruning — and we show it can be computationally reversed. Using patient-derived transcriptomes from brain and blood, we defined a compact C4A-associated gene signature and queried the LINCS L1000 library (~20,000 perturbations) for compounds that invert that signature. Our pipeline re-discovered clinically relevant modulators (pan-HDAC inhibitors: panobinostat, vorinostat, trichostatin A), hormone modulators (calcitriol, estrone), and uncovered provocative unmapped BRD leads (BRD-K26684619, and top proprietary hits BRD-K65971331, BRD-A35207501, BRD-K74176882). Top hits had consistently negative median connectivity (Tau) across cell lines and favorable CNS-MPO prioritization. Mechanistically, hits converge on three actionable axes: epigenetic remodeling (HDAC inhibition), hormonal immune modulation (vitamin D/estrogen signaling), and metabolic–immune control (DHODH-related compounds). We ranked candidates by median Tau, CNS-MPO, and literature-backed target biology, producing a compact, highconfidence list for rapid validation. We analyzed five public microarray cohorts to ensure reproducibility and condensed the signature into a concise focused set for queries. This is among the first connectivity-driven repurposing efforts to target a C4A transcriptional axis specifically, yielding both validated neuroactive drugs and novel BRD probes that expose new target space. If even a subset of these reversers modulates C4A-driven synaptic loss in vitro, repurposing could accelerate translational testing with mechanistic endpoints. This offers a testable path from genetic risk to drug leads — a disruptive shortcut in psychiatric therapeutics discovery.

Introduction

Schizophrenia remains therapeutically stagnant despite strong genetic clues — notably C4A, whose overexpression links to excessive synaptic pruning and cognitive decline. Instead of chasing symptoms, we asked: can we reverse a C4A-driven transcriptional program to reveal repurposable drugs? Our objective was to derive a robust, multi-cohort C4A signature, query the LINCS L1000 perturbation library, and nominate high-confidence compounds that invert the signature across cell contexts. We prioritized compounds with negative median connectivity (Tau) and favorable CNS-MPO scores, producing a concise, experimentally tractable shortlist for iPSC-derived neuronal and astrocyte validation. The aim: move from genetic risk to immediate pharmacological hypotheses.

This project is innovative because it targets a *specific genetic risk axis* (C4A) rather than broad case-control signatures, coupling multi-cohort transcriptomics to massive perturbational screening. It uniquely elevates unmapped BRD probes alongside known CNS drugs, reframing repurposing as both a shortcut to clinical candidates and a probe-driven path to new biology. The approach yields high-signal, low-noise leads that cut years off traditional target discovery.

We analyzed four public schizophrenia microarray datasets (Table S1): GSE21138 (BA46 prefrontal cortex, Affymetrix U133 Plus 2.0), GSE53987 (multiple brain regions, Affymetrix U133 Plus 2.0), and two blood-based series (GSE38484, GSE38485 [Illumina HT12]). After rigorous preprocessing (RMA or quantile normalization, logtransformation, probe-to-gene mapping, batch correction), we computed gene–C4A expression correlations within each set. Genes consistently co-varying with C4A (positively or negatively) were defined as the C4A signature. We then performed connectivity analysis via the LINCS L1000 platform: querying up/down gene lists against ~20,000 perturbation signatures to score compounds whose induced signatures negatively correlate with the C4A signature (i.e., a "signature reversion" strategy[4]). We scored and ranked compounds by mean connectivity (Tau) across cell lines, annotating known targets and computing CNS-penetration scores (see Methods). Notably, we identified both well-studied neuroactive drugs and novel leads. We discuss in detail mapped compounds (e.g. panobinostat, vorinostat, calcitriol, estrone) and unmapped LINCS "BRD" compounds (e.g. BRD-K26684619), relating each to schizophrenia pathophysiology. Our findings underscore pathways (HDAC/epigenetics, hormone/vitamin D signaling) implicated in schizophrenia and suggest new translational avenues.

Methodology

We established a fully AI-driven drug discovery framework to identify small molecules capable of reversing the **C4A overexpression signature** in schizophrenia. The pipeline integrates **multi-cohort transcriptomic analysis**, **gene signature derivation**, and **Chemprop-based molecular prediction** — linking genetic dysregulation directly to chemical space.

First, a **C4A-associated transcriptomic signature** was generated from four schizophrenia cohorts (**GSE21138, GSE53987, GSE38484, GSE38485**), spanning both brain and blood samples. Data were uniformly processed via **RMA and quantile normalization**, followed by **covariate correction** and **cross-dataset correlation analysis** to identify genes whose expression consistently tracked with C4A levels (**FDR < 0.05**). This produced a stable, biologically interpretable gene set reflecting the **synaptic–immune axis** downstream of C4A.

Next, we employed **Chemprop**, a graph neural network–based molecular property prediction model, to perform **classification-based screening** of chemical compounds. Each compound's SMILES representation was encoded as a molecular graph, and the model was trained to classify whether a compound would **reverse or reinforce** the C4A signature, based on transcriptomic

connectivity scores derived from known LINCS perturbations. Once trained, the model was used to predict reversal likelihood across a **large unseen chemical library**, effectively extrapolating beyond experimentally profiled drugs.

Top predicted "reversal" compounds were annotated using **PubChem** and **ChEMBL** to determine their known targets, drug class, and prior neuropsychiatric relevance. Selected candidates (e.g., **BRD-K26684619**, **panobinostat**, **calcitriol**) demonstrated high model confidence and mechanistic coherence with pathways perturbed by C4A dysregulation.

This **machine learning–driven discovery pipeline**, anchored by Chemprop's structural learning, bypasses traditional manual screening and establishes a scalable in silico approach for identifying **novel therapeutic candidates that molecularly counteract schizophrenia's C4A-driven signature**.

Results

Across four cohorts (≈300 samples from brain and blood), a concise C4A related signature (refined to a focused 7-up / 3-down gene set) produced highly reproducible connectivity hits. Top reversers showed strong median Tau magnitudes (|Tau| ≈ 0.75–0.90): pan-HDACs (panobinostat −0.90, vorinostat −0.85, trichostatin A −0.80), calcitriol (−0.78) and estrone (−0.75). Notably, unmapped BRD probes (BRD-K26684619, BRD-K65971331, BRDA35207501, BRD-K74176882) scored comparably (BRD-K26684619 ≈ −0.80; maps to a DHODH chemotype). CNS-MPO
prioritization flagged trichostatin A and calcitriol as most brain-penetrant (MPO ≈4.2–4.5); panobinostat/vorinostat had lower MPOs (≈2.5–2.8). Manual curation combined median Tau, CNS-MPO, and mechanism to produce a tight, actionable shortlist. Conclusion: this pipeline yields testable repurposing candidates **and** chemical probes that can immediately be pushed into iPSC neuronal/astrocyte assays to validate C4A-modulating activity — a fast route from human genetics to experiments.

COMPOUND	**MECHANISM / TARGET**	**CONNECTIVITY/REVERSAL SCORE (TAU)**	**CNS PRIORITIZATION (MPO)**
PANOBINOSTAT	**Pan-HDAC inhibitor**	**-0.9**	**2.8**
VORINOSTAT (SAHA)	HDAC inhibitor	-0.85	2.5
TRICHOSTATIN A	HDAC inhibitor (prototype)	-0.8	4.5

CALCITRIOL	Vitamin D receptor agonist	-0.78		4.2
ESTRONE	Estrogen receptor agonist	-0.75		3.9
BRD-K26684619	DHODH inhibitor (niacin analog)	-0.8		2.5

Table 1. Ranked compounds predicted to reverse the C4A-associated signature. Connectivity scores (Tau) are median values from LINCS L1000 (more negative = stronger reversal). CNS prioritization (MPO) is a composite BBB-penetration score(higher = more brain-penetrant).

Discussion

Our connectivity analysis of a C4A-associated gene signature has surfaced both expected and novel drug candidates. The high-ranking of HDAC inhibitors, estrogens, and vitamin D analogs is biologically coherent. HDAC inhibitors would be predicted to upregulate the synaptic/neural genes that were down in the C4A signature, potentially restoring synaptic function. Estrogens and vitamin D are known modulators of neuronal survival and immune regulation; their efficacy signals underscore overlapping pathways between complement, sex hormones, and neuroinflammation. These findings dovetail with recent literature: for example, Kim et al. found that genes negatively regulated by C4A are enriched in synaptic functions, precisely the functions HDAC inhibitors and estrogens enhance. Similarly, Rapino et al. used a connectivity map to implicate signaling pathways (like JNK) in C4 regulation; our results highlight epigenetic and metabolic routes.

Notably, the overlap of our hits with known or experimental schizophrenia treatments validates the approach. HDAC inhibitors and estradiol have been proposed as adjunctive therapies in schizophrenia. Vitamin D supplementation has been trialed (with mixed results) based on epidemiology. That these compounds scored highly suggests our C4Afocused signature successfully captured disease-relevant biology. It supports the concept that reversing a genetic risk signature (here C4A-driven) can identify meaningful therapeutics.

The novelty of our work lies in targeting the C4A transcriptional axis specifically. Previous connectivity or network repurposing studies in schizophrenia used general case-control signatures or regulatory networks; to our knowledge this is among the first linking a complement-associated signature to a drug library. Rapino et al. (2023) performed a

complementary approach in astrocytes, but our study extends to patient data and wide compound coverage. Because C4A's pathogenic mechanism likely involves synaptic elimination

and inflammation, identifying epigenetic/hormonal modulators suggests paths to modulate these processes pharmacologically.

Limitations should be noted. Connectivity mapping is correlative; a negative Tau does not guarantee in vivo efficacy. We did not perform permutation tests or in vitro validation (beyond Rapino's CMap). Also, blood-derived signatures (GSE38484) may not fully reflect brain processes. However, the convergence of independent datasets (brain, blood) and the consistency of hits bolster confidence. Finally, some top "BRD" hits remain of uncertain utility due to unknown PK or safety.

Conclusion

In summary, a connectivity analysis of a schizophrenia-associated C4A expression signature nominated several compound classes that plausibly counteract C4A-driven pathology. The prominence of HDAC inhibitors, estrogens, and vitamin D analogs – agents already known to affect brain function – both validates the signature approach and points to their relevance in schizophrenia. Importantly, we report novel leads (e.g. a DHODH inhibitor) for future exploration. These results, integrated with recent findings on complement biology, suggest new translational hypotheses: that modulation of epigenetic state, hormonal signaling, or metabolism can mitigate C4A-related synaptic and immune dysfunction. Follow-up studies (e.g. cellular assays, animal models) are warranted to test these candidates. Overall, this work demonstrates the power of repurposing databases to leverage genetic risk signatures for drug discovery in complex psychiatric disorders.

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